SHIV NADAR UNIVERSITY

CSD 301

Software Engineering

FINAL PROJECT REPORT

FoodOn Web-App

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Acknowledgement

We'd like to take this opportunity to extend our heartfelt gratitude
to the university and Professor Pooja Malik for the wonderful opportunity
to design and deploy our own software from scratch.

We truly enjoyed the course and what it taught us,
and hope to see this project in action by early next year.

We would like to thank ma'am for her ceaseless support and guidance throughout
the course of the semester, and for helping us with this project.

We would also like to thank FoodOn for the idea and access to their menus.

Problem Statement

1. Project Background and Description

FoodOn is an on-campus startup that works on the idea of delivering food from restaurants that don't deliver to SNU at present. They take orders from students using Google Forms, collect individual payments and go to the restaurant to pick up the collective order. The delivery happens weekly, and while it enjoys tremendous success on campus, the present method of ordering and payment collection is haphazard to say the least.

2. Project Scope

Through this project, we aim to build a web-app for FoodOn to use. It will be a live project that allows user to place orders in a much easier fashion, pay at their own convenience and get updates on order status. We will also try to incorporate live order tracking as an additional feature. A loyalty program and points system will be introduced to incentivize customers to order regularly.

3. High-Level Requirements

FoodOn users will have the ability to:

- Create user accounts and log in.
- Update personal and contact information.
- View the menu that has been set visible for the day.
- Place orders on the website.
- Pay via the website using a payment gateway/QR code.
- Monitor order history and repeat orders.
- View upcoming restaurant schedule.
- Redeem their reward points on subsequent orders.
- Track their orders in real-time.

FoodOn admins will have the ability to:

- Set the menu for the day.
- Update the upcoming restaurant schedule.
- Put out updates for all the customers.
- Monitor user accounts.
- Collect payments.
- Open and close the order window and payment portal.
- Suspend user accounts if necessary.
- Initiate refunds via the website.
- Monitor P/L on a daily basis.

Software Requirement Specifications (SRS)

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1. Introduction

The FoodOn Web-App version 1.0 is to be developed for SNU students and staff. FoodOn is an on-campus startup that works on the idea of delivering food from restaurants that do not deliver to SNU at present. They take orders from students using Google Forms, collect individual payments and go to the restaurant to pick up the collective order. The delivery happens weekly, and while it enjoys tremendous success on campus, the present method of ordering and payment collection is haphazard to say the least.

1.1 Purpose

This SRS defines the User Interface, Frontend and Backend of the FoodOn Web-App version1.0.
This document is intended for the following group of people:-

- *Developers for the purpose of maintenance and new releases of the software.*
- *FoodOn Administrators*
- *Documentation writers.*
- *Testers.*

1.2 Scope

This document applies to FoodOn Web-App version 1.0. Through this software, we aim to build a web-app for FoodOn to use. It will be a live project that allows user to place orders in a much easier fashion, pay at their own convenience and get updates on order status. We will also try to incorporate live order tracking as an additional feature. A loyalty program and points system will be introduced to incentivize customers to order regularly.

1.3 Definitions, Acronyms and Abbreviations

FoodOn	*It is an on-campus startup that works on the idea of delivering food from restaurants*
GPS	*Global Positioning System*
RP	*Reward Points*
SNU	*Shiv Nadar University*
Web-app	*Web Application*

1.4 Overview

- ***Section 1.0*** *Discusses the purpose and scope of the software*
- ***Section 2.0*** *Describes the overall functionalities and constraints of the software and the user characteristics*
- ***Section 3.0*** *Details all the requirements needed to design the software*
- ***Section 4.0*** *Describes the features provided by the system software*

2. The Overall Description

2.1 Product Perspective

- *The FoodOn web-app consists primarily of two functional components which consist various subcomponents.*
- *One of the Functional component deals with the admin side and the other deals with customer interactions.*
- *The admin component allows the admin to check order history, to select and update different menus, to put out notices for the customers etc.*
- *The user component allows the customer to login with their gmail ID to place orders, to select payment method, to see upcoming schedule of restaurants, to see different restaurant menus.*
- *This software uses a Cloud Database and therefore does not require any server to host the web-app.*

2.2 Product Functions

The major functions that the FoodOn web-app will perform are –

User Side

- ***Placing Order** – The user will be able to add items to a cart from a wide variety of menus made available to them by the admin. He/she can then confirm their order after selecting a payment method.*
- ***View Restaurant Schedule** – The user will be able to view the schedule of the restaurants in the upcoming weeks.*
- ***Redeem Reward Points** – The user will be able to redeem and earn reward points.*
- ***Order History** - Any transaction shall be recorded in the database. The customer can view his/her history from the user page.*
- ***Cancelling** – The customer will be able to cancel their order if permitted my the admin.*
- ***Live order tracking** - Users will be able to track their orders.*

Admin Side

- ***Creating and Updating Menus** - The admin will be able to create new menus and modify existing ones.*
- ***Setting Menu** – The admin will be able to set the menu for the day i.e the menu from which the users can place order for the day.*
- ***Announcements** – The admin will be able to put out notices which every customer will be able to see.*
- ***Monitor Order History** – The admin will be able to see and monitor all the orders placed and also the order history of different customers.*
- ***Initiate Refunds** – If a user has cancelled their order, then the admin will be able to refund the payment through the application.*

2.3 User Characteristics

The different users which will be using tis software are –

- ***Students*** *– The main FoodOn customer base are the students of SNU and they will be the primary users of this application.*
- ***SNU Employees*** *– A small percentage of SNU employees are also part of the FoodOn Consumer base.*
- ***Admin*** *– The FoodOn employees are the only ones to have access to the admin page. He/she has complete access to the software.*

2.4 Constraints

- ***Database Capacity***
  The capacity of the database is not infinite. Eventually , there will come a time when the database is full and old records of the customer have to be deleted to make room for the new records.

3. External Interface Requirements

3.1 User Interface Requirements

- *A navigation bar will be there on top of the page, which will allow you to login as a customer or as an admin. It will also contain a button which displays the schedule of restaurants.*
- *The different restaurant logos will be shown on the main page and clicking on any of them will allow to you look at their menu and to add items to your cart.*
- *After logging in as the user you will be allowed to look at your cart and will be able to confirm your order.*
- *After clicking on admin login you will be taken to a new page to login as the admin.*
- *After successfully logging in as the admin, a screen will be shown having the options to set the menu, to make an announcement and to view the orders.*

3.2 Hardware Interface Requirements

- *The user should just have a mobile device or a laptop with an active internet connection.*
- *The person delivering the food should have a device which enables us to track his/her live location.*

4. System Features

4. 1 Safe and secure payment portal

A safe and secure payment portal will be created which would allow users to pay through commonly used payment gateways without fearing the loss of data. Any passwords used will be encrypted.

4.2 Precise Live Order Tracking

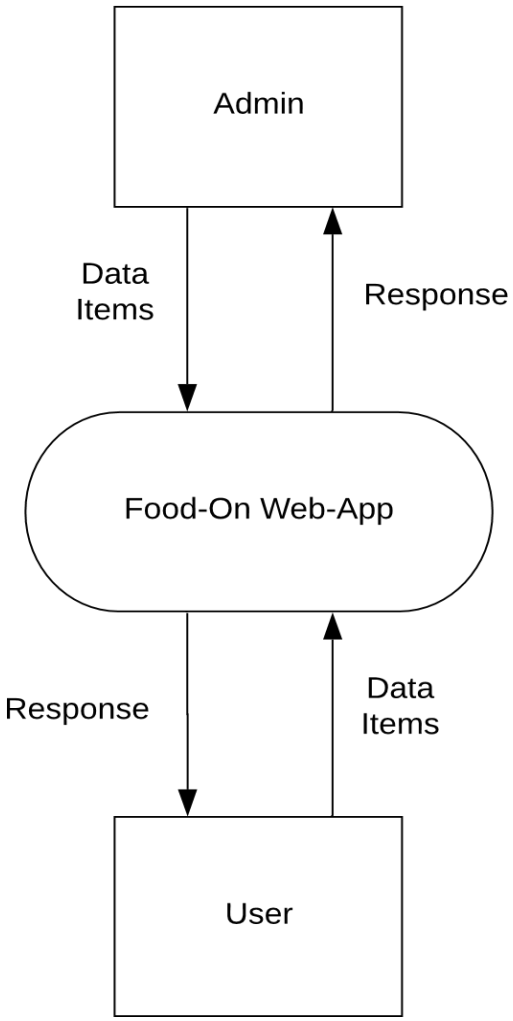
Live order tracking will be implemented using latest GPS technology. This would help customers get precise and accurate location of the food to be delivered.

4.3 Admin and user validation

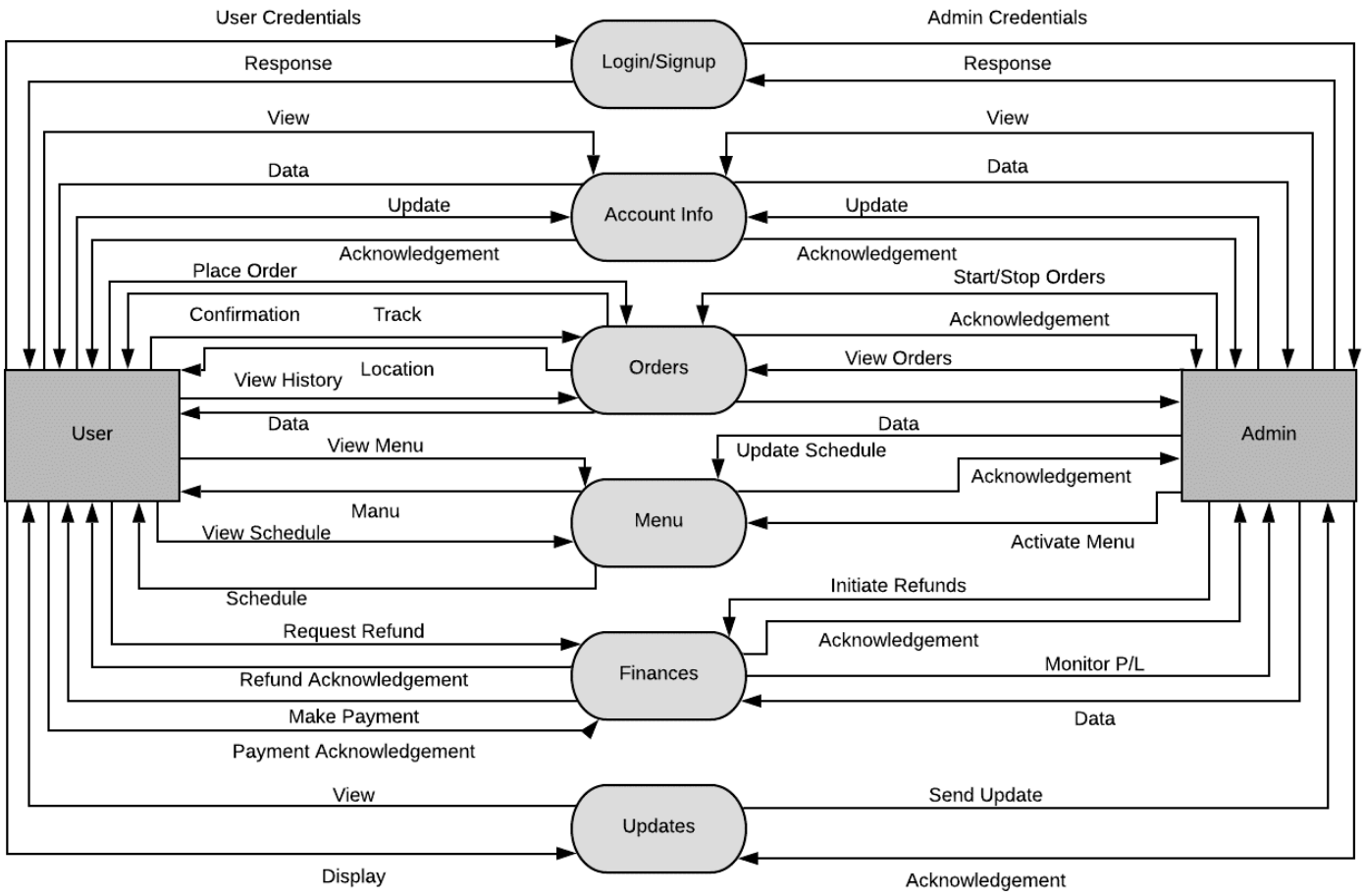
Secure and strict admin and user validation. A single user can be active through only one device at a time. Passwords are encrypted.

Data Flow Diagrams

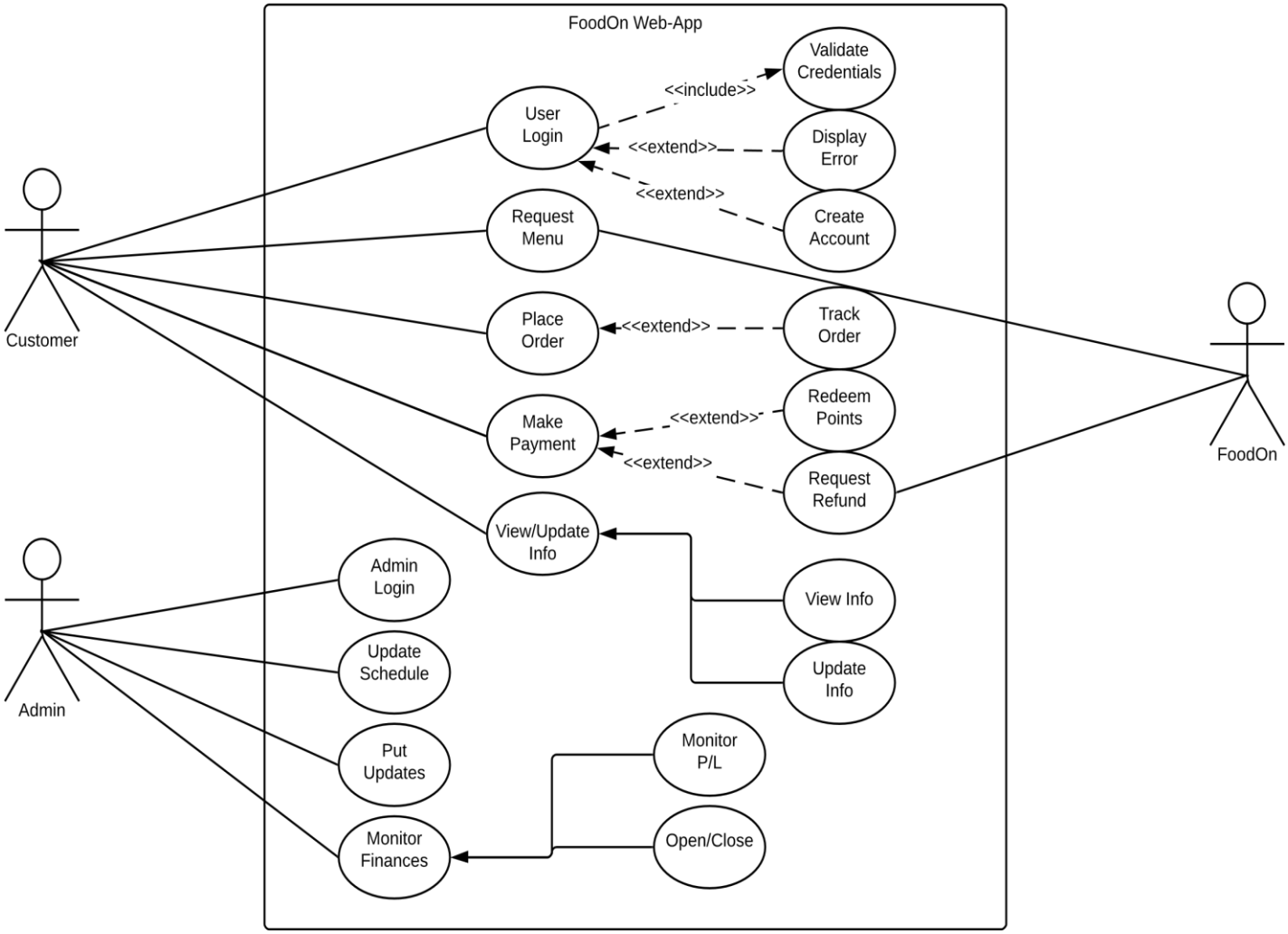
Level 0



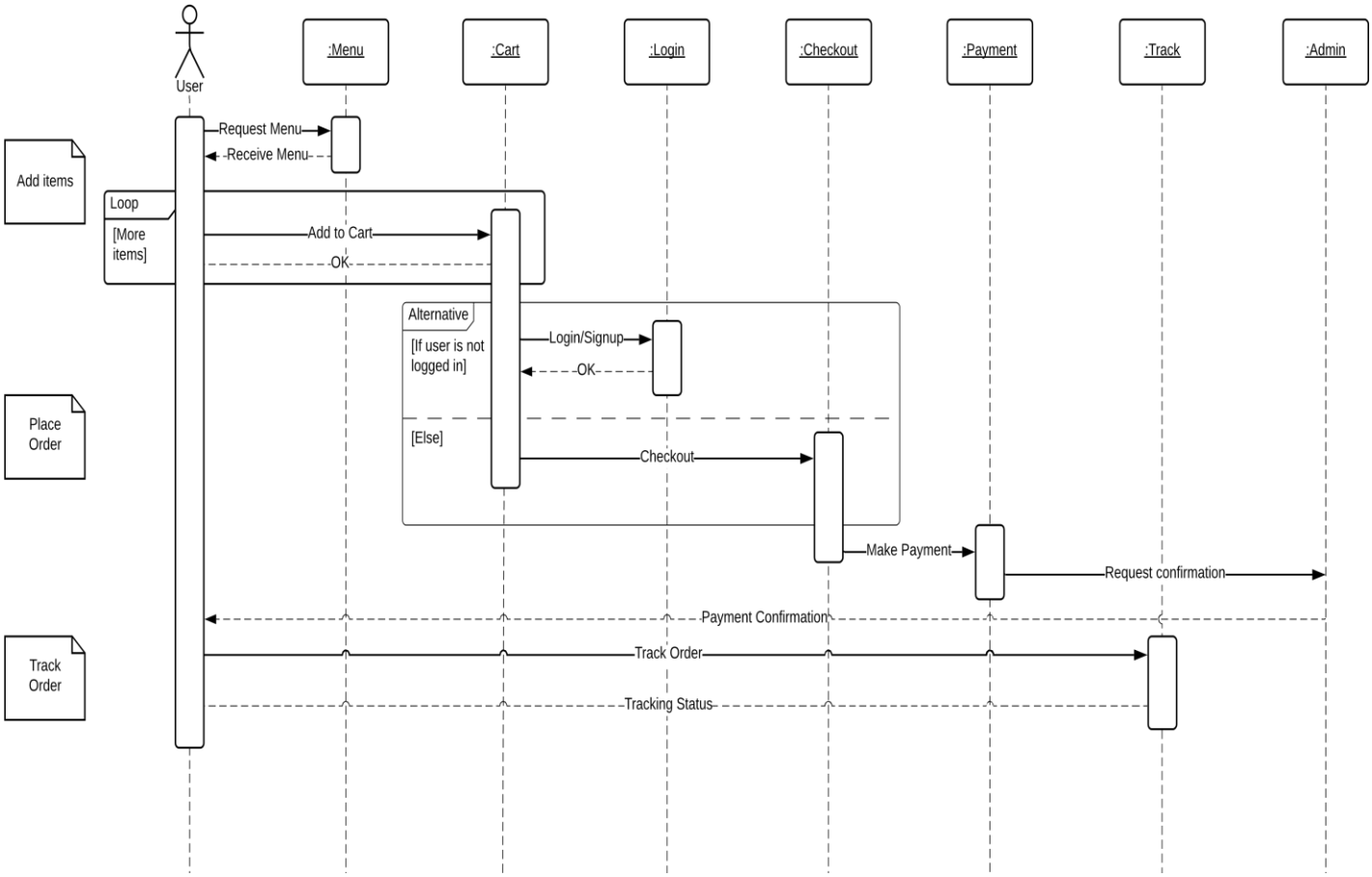
Level 1


User Credentials
Admin Credentials
Login/Signup
Response
Response
View
View
Data
Data
Account Info
Update
Update
Acknowledgement
Acknowledgement
Place Order
Start/Stop Orders
Confirmation
Track
Acknowledgement
Orders
View Orders
Location
View History
User
Admin
Data
Data
View Menu
Update Schedule
Acknowledgement
Menu
Manu
View Schedule
Activate Menu
Schedule
Initiate Refunds
Request Refund
Acknowledgement
Finances
Monitor P/L
Refund Acknowledgement
Make Payment
Data
Payment Acknowledgement
View
Send Update
Updates
Display
Acknowledgement

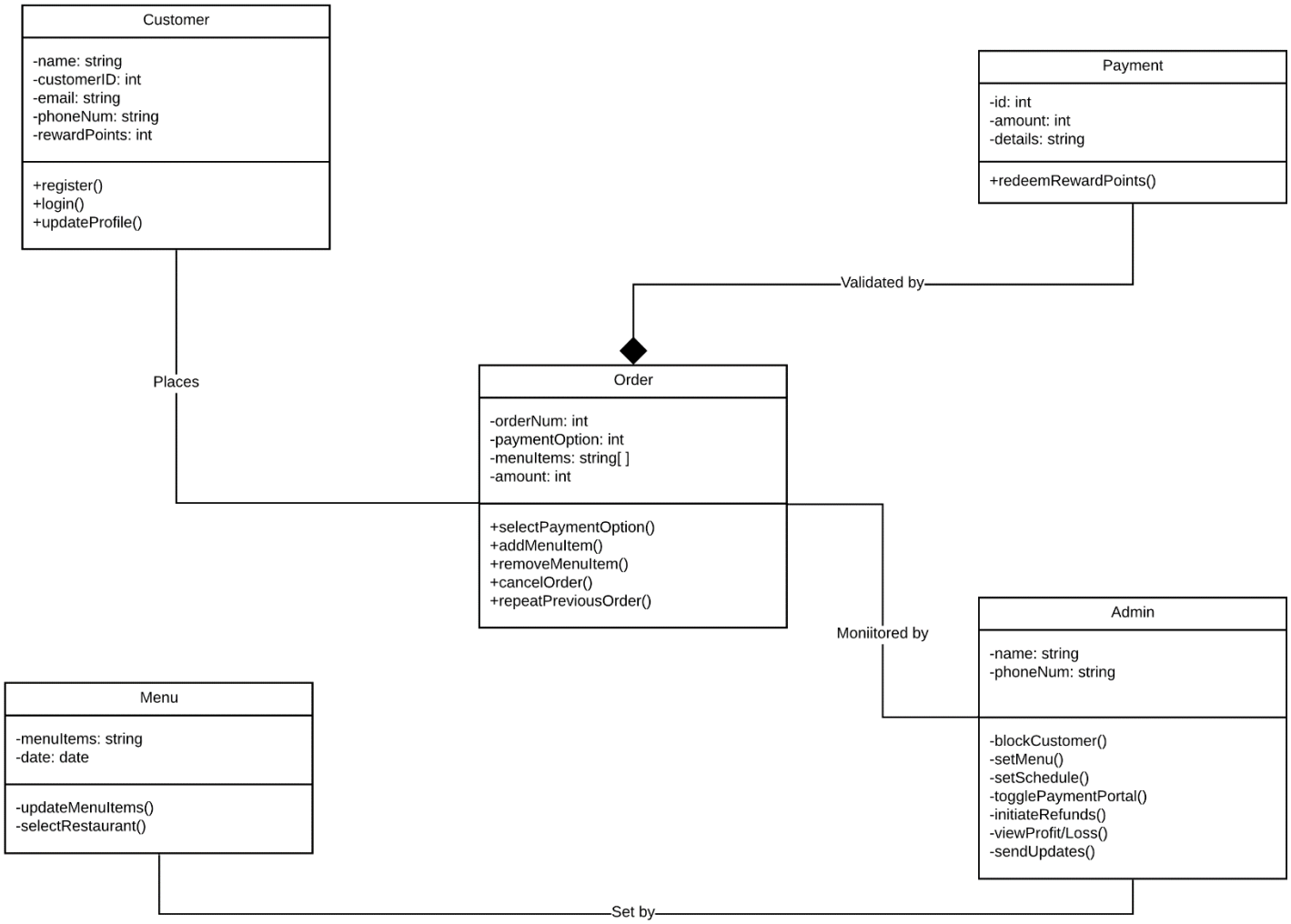
Use-Case Diagram



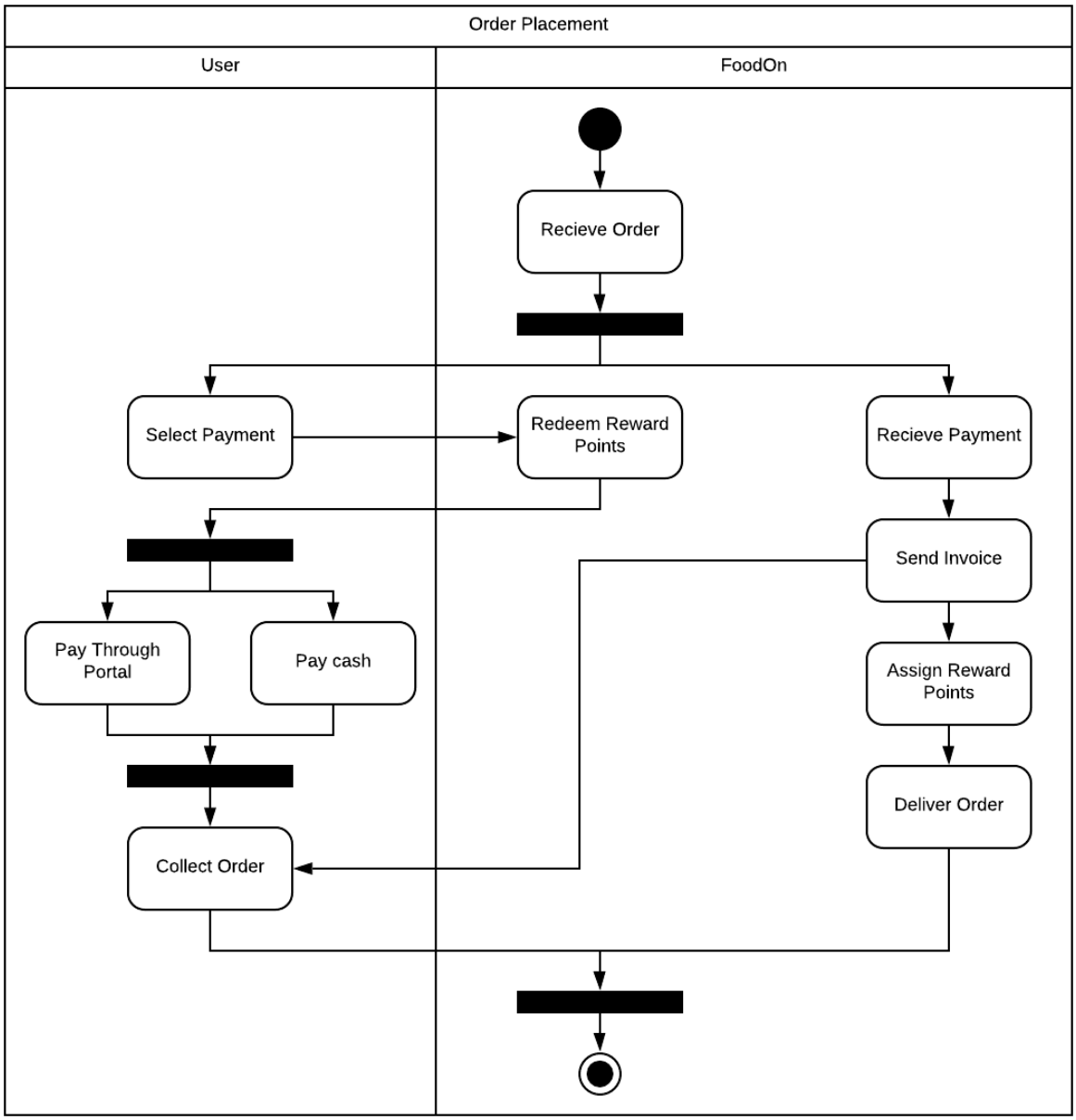
Sequence Diagram



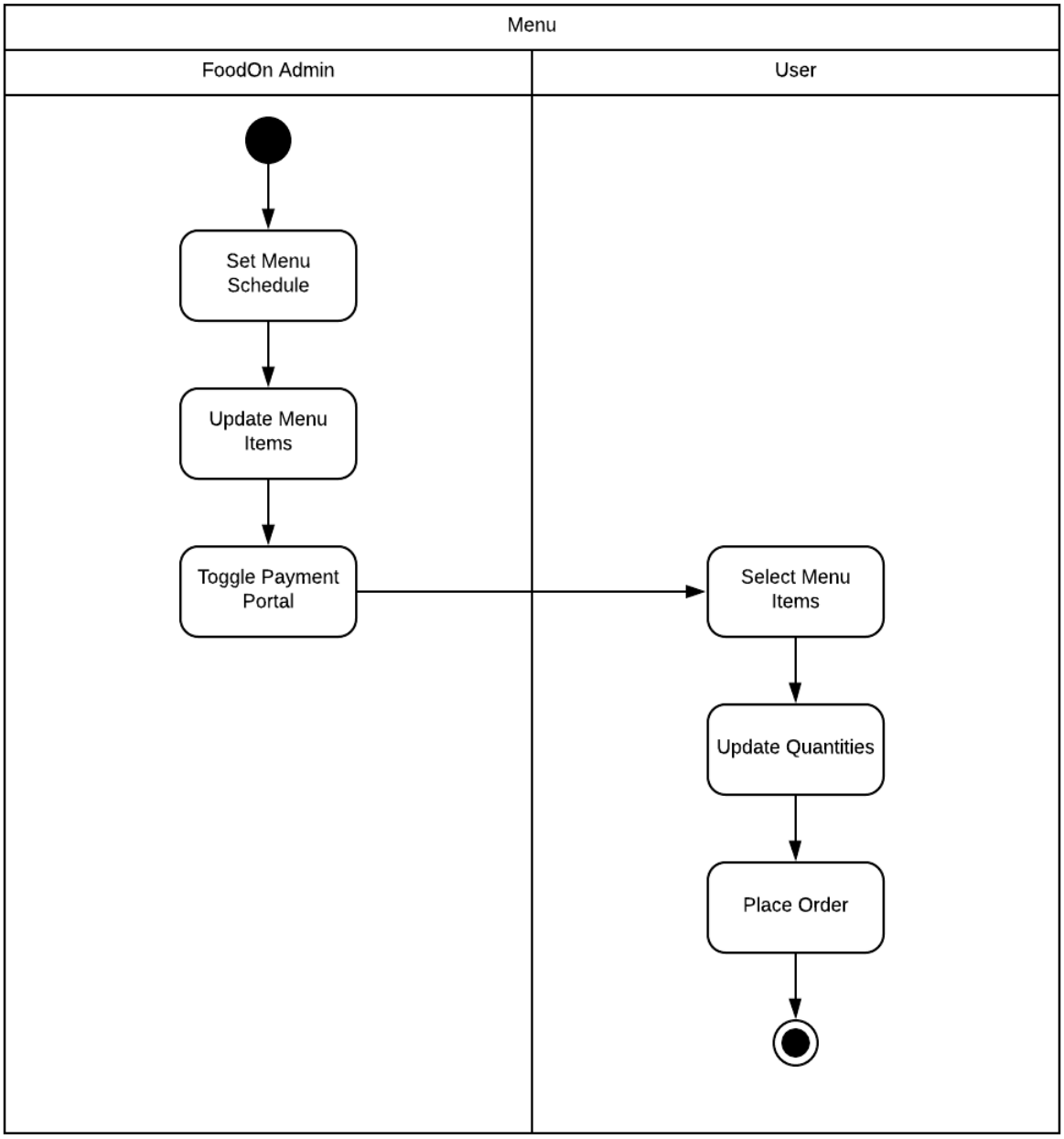
Class Diagram



Activity Diagrams

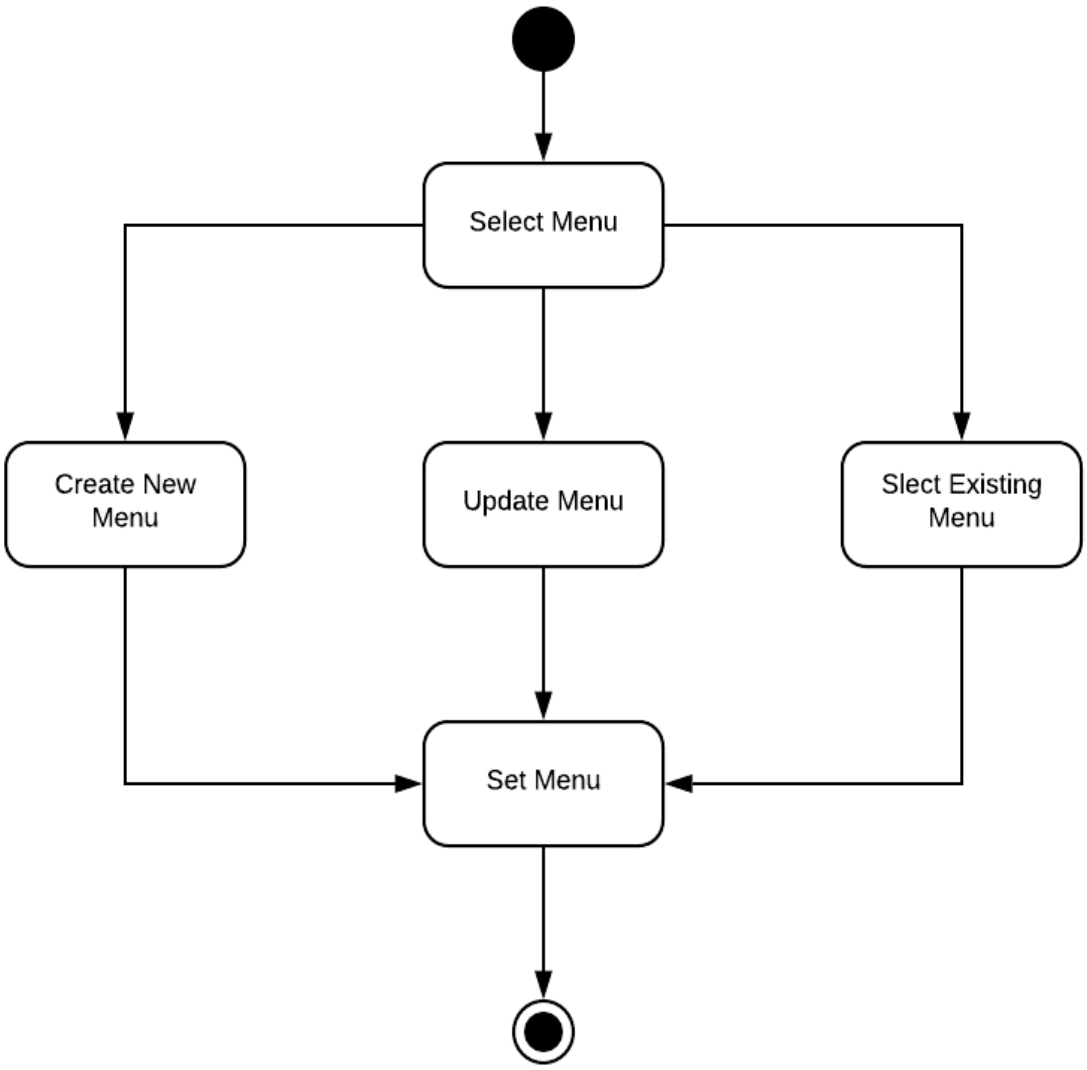


Retrieve Order

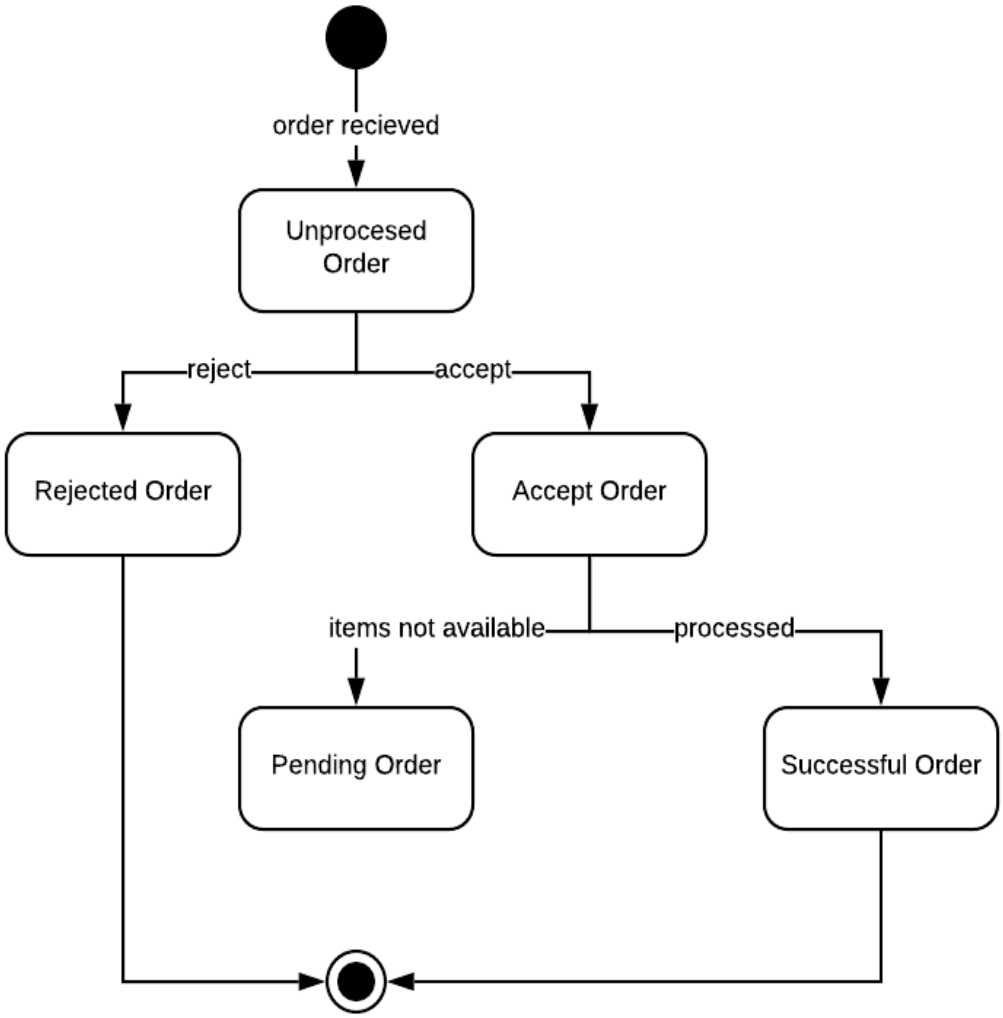

Menu
FoodOn Admin
User
Set Menu Schedule
Update Menu Items
Toggle Payment Portal
Select Menu Items
Update Quantities
Place Order

Set Menu

State Chart Diagrams



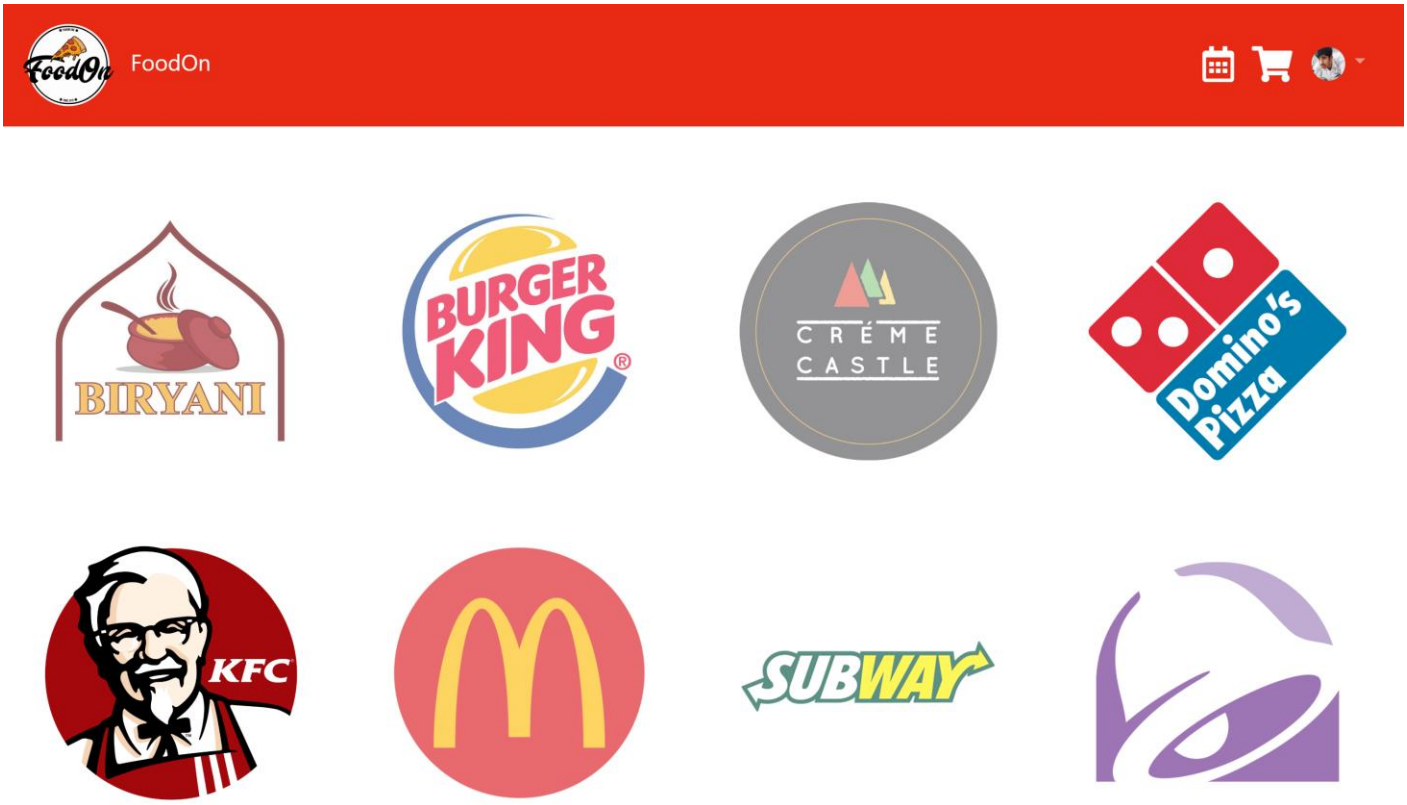
Menu



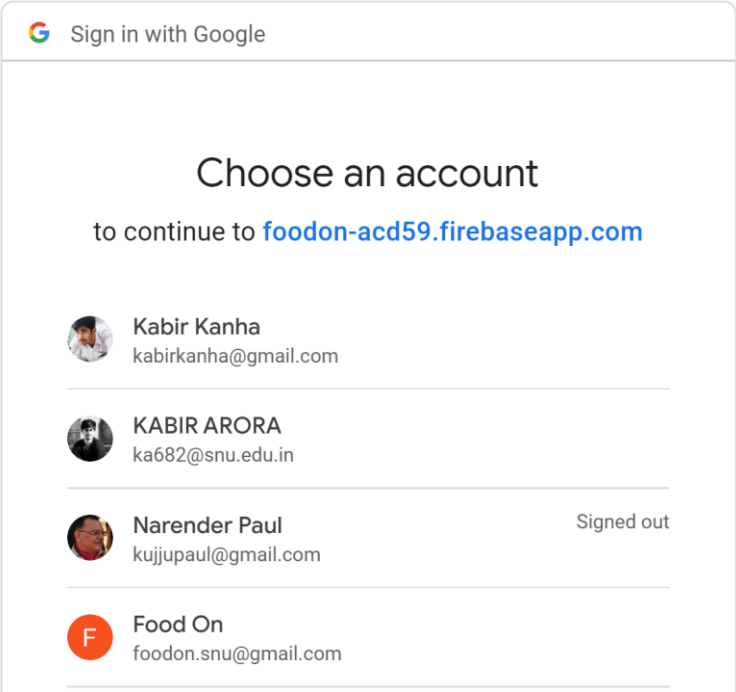
Placing Order

Web-App

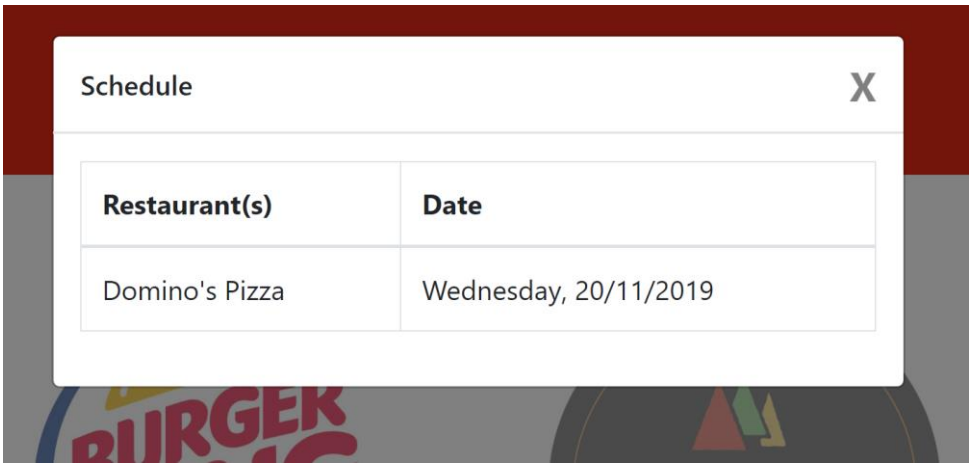
Some screengrabs from our web-app:



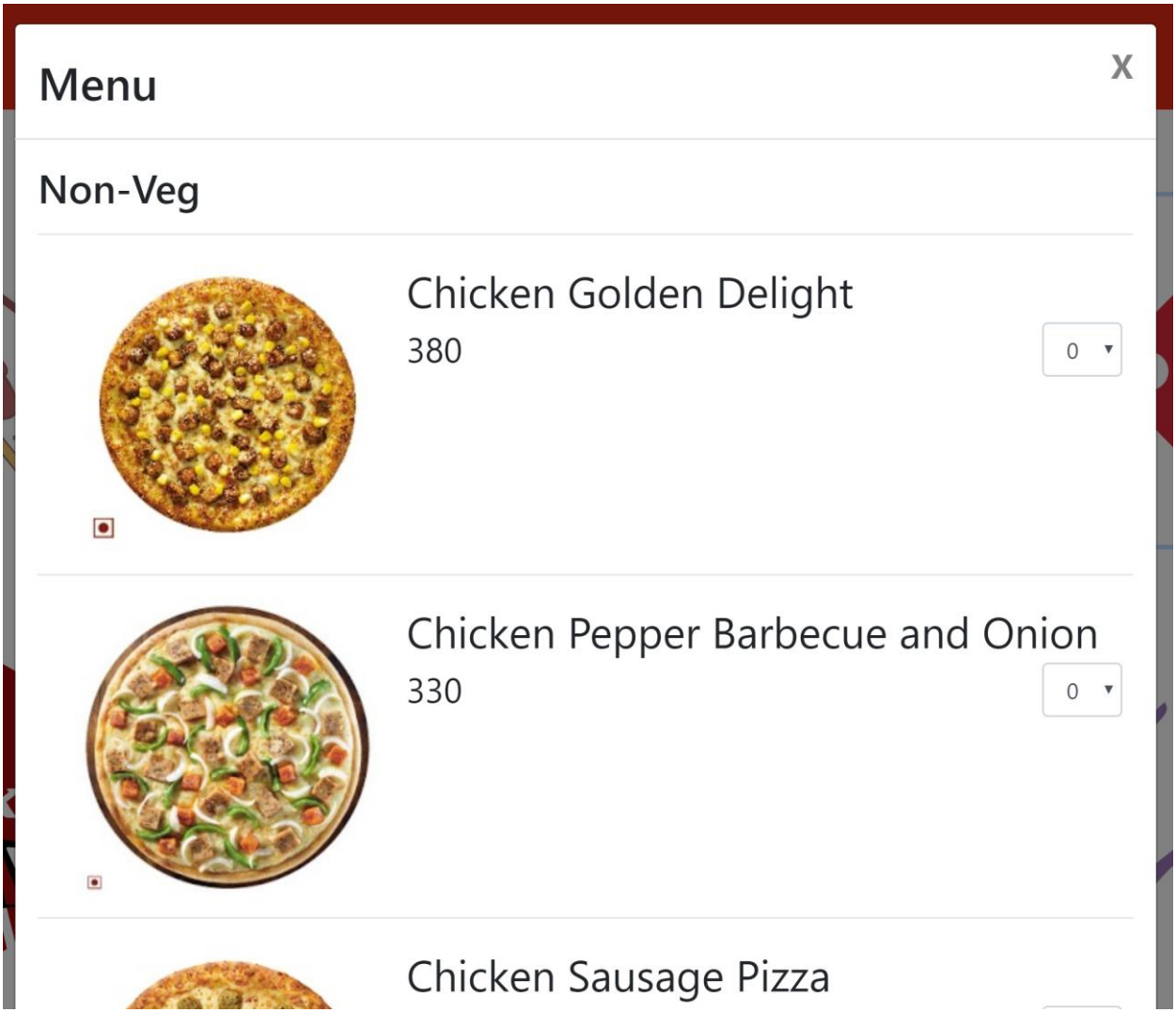
Main Page



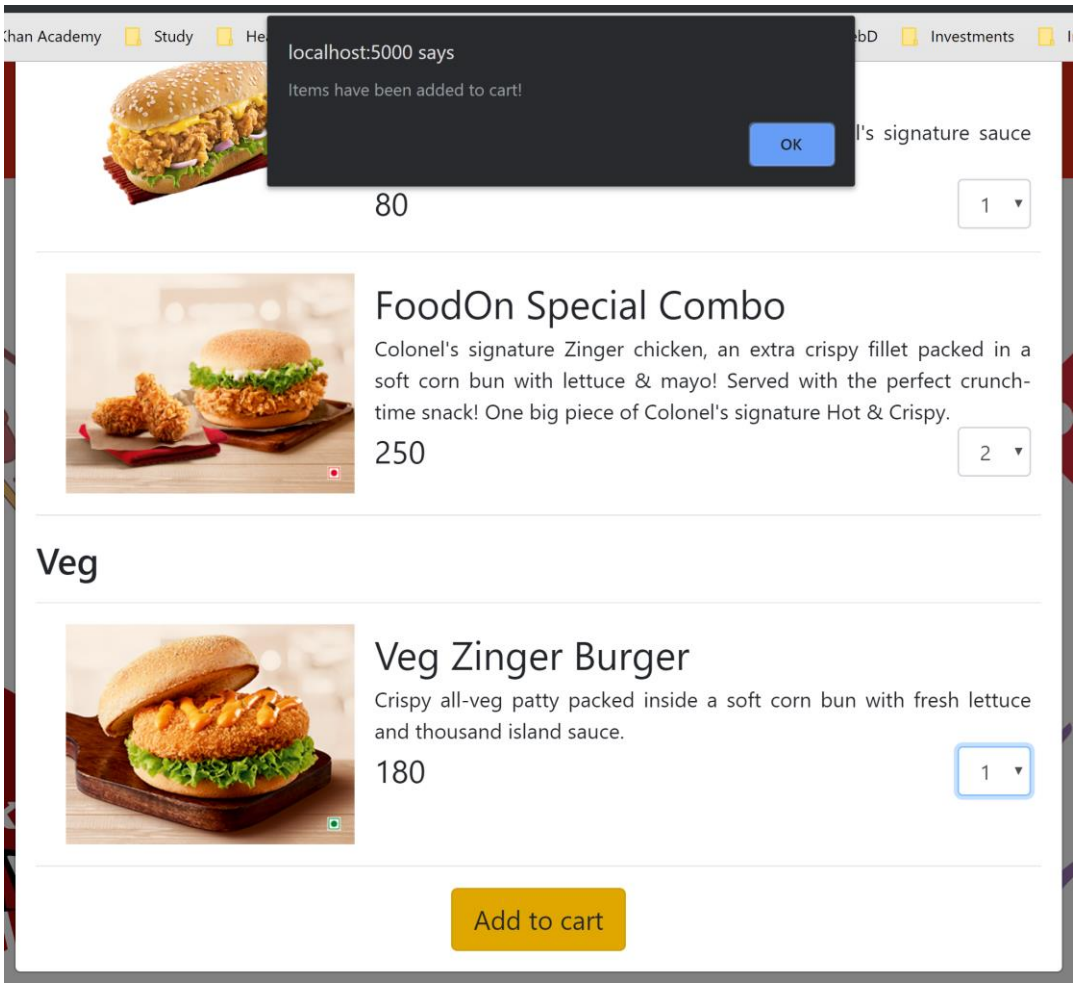
Sign-In with Google



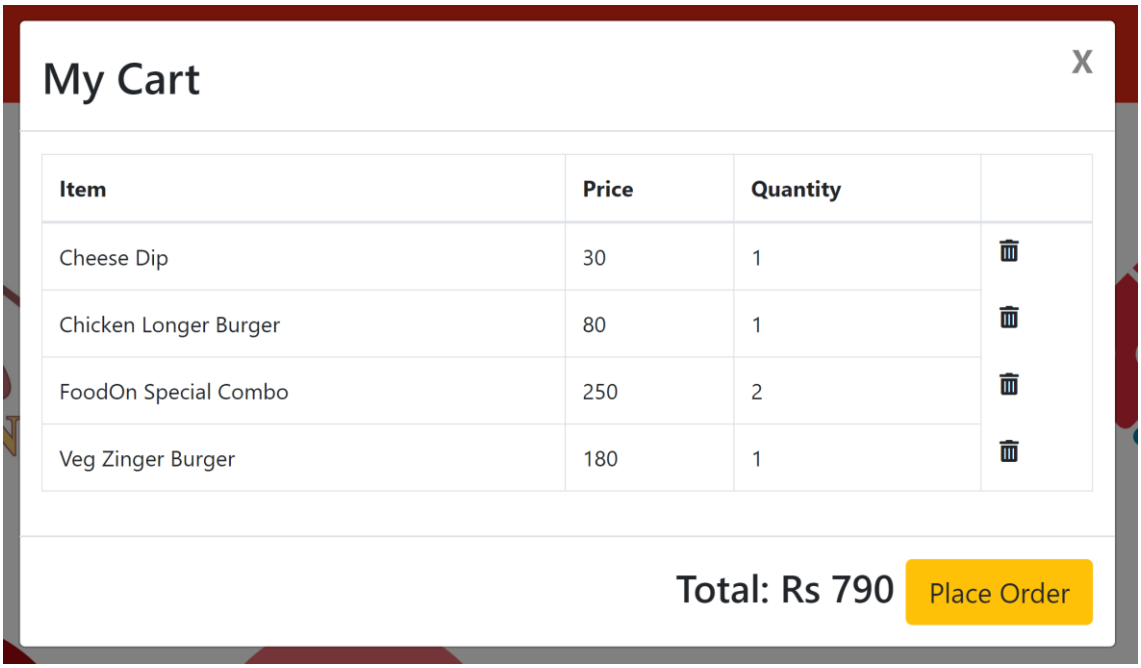
Upcoming Schedule



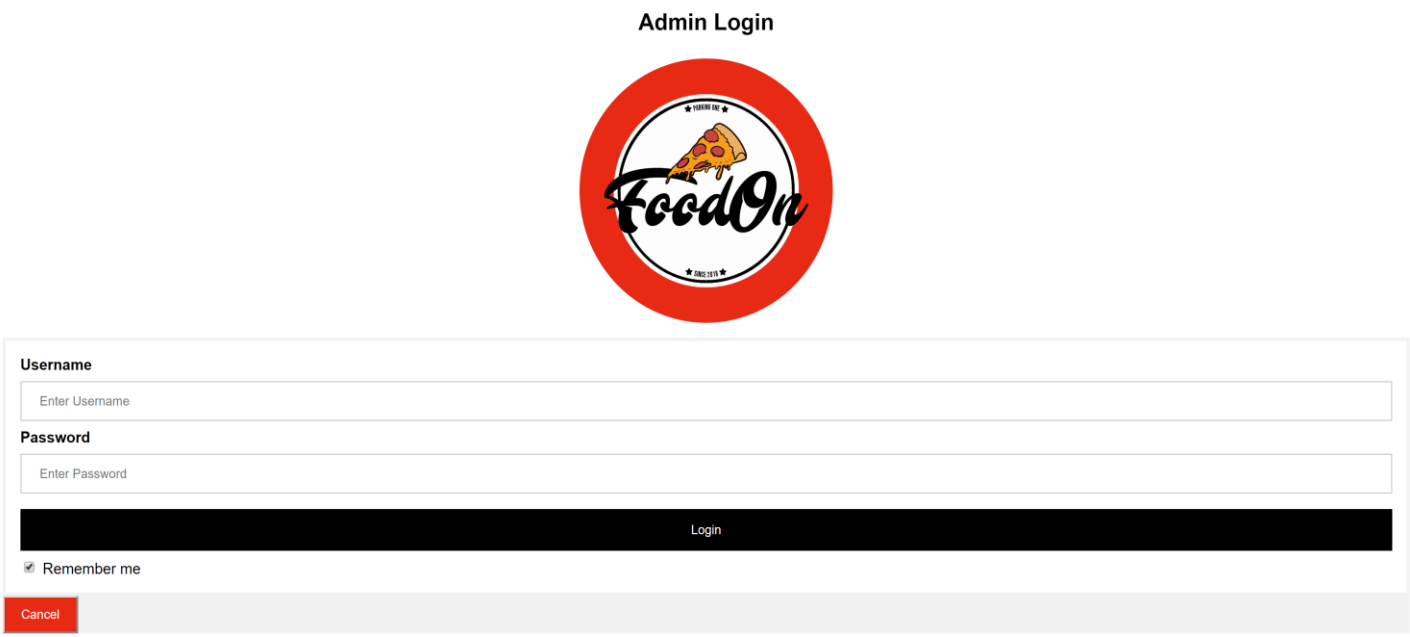
Building the order



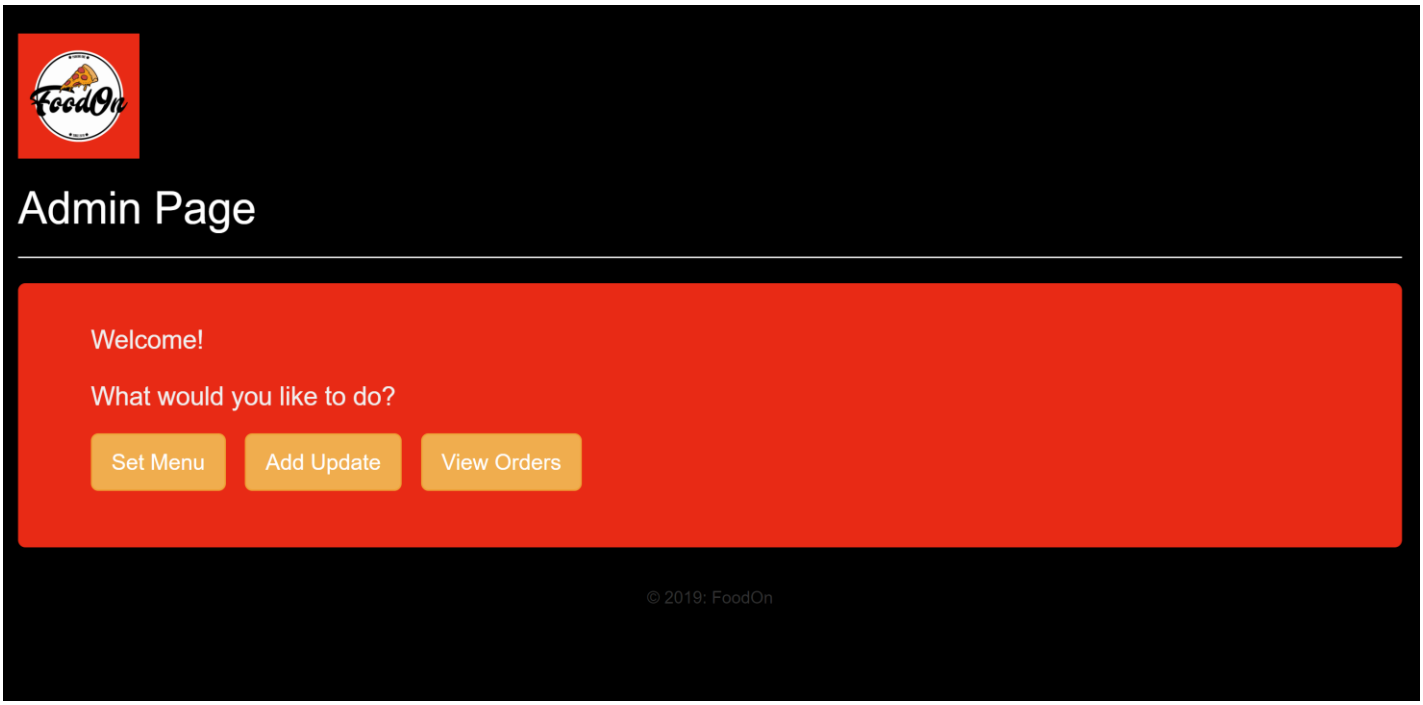
Adding items to cart



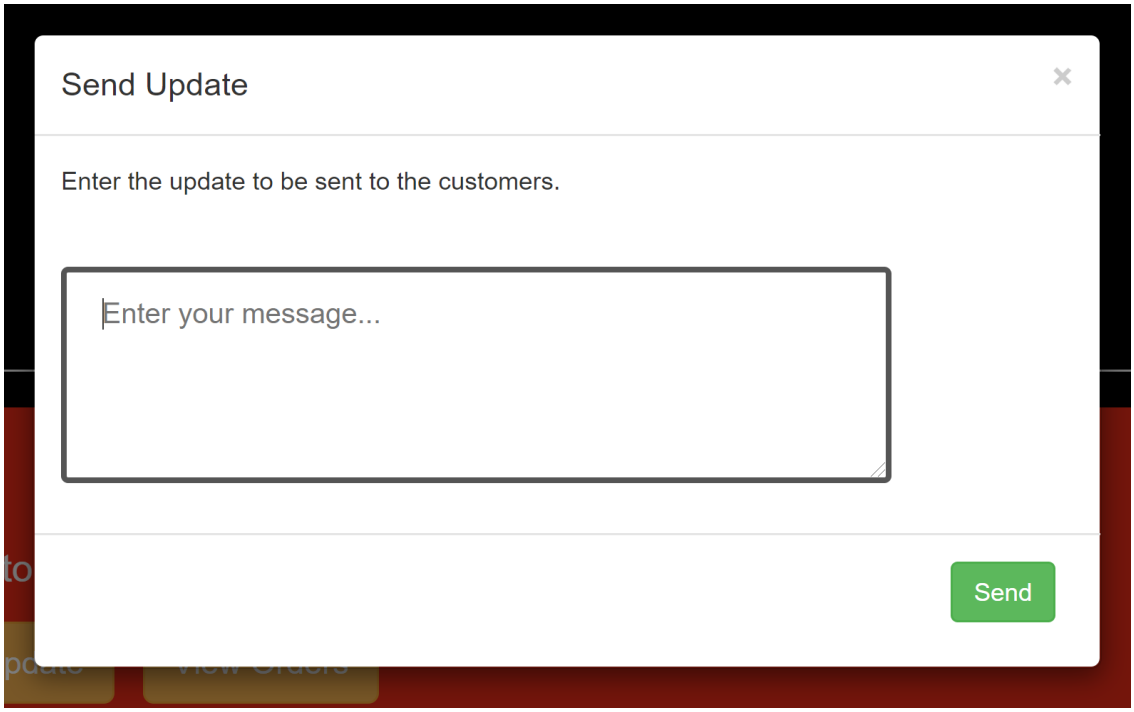
Placing order from cart



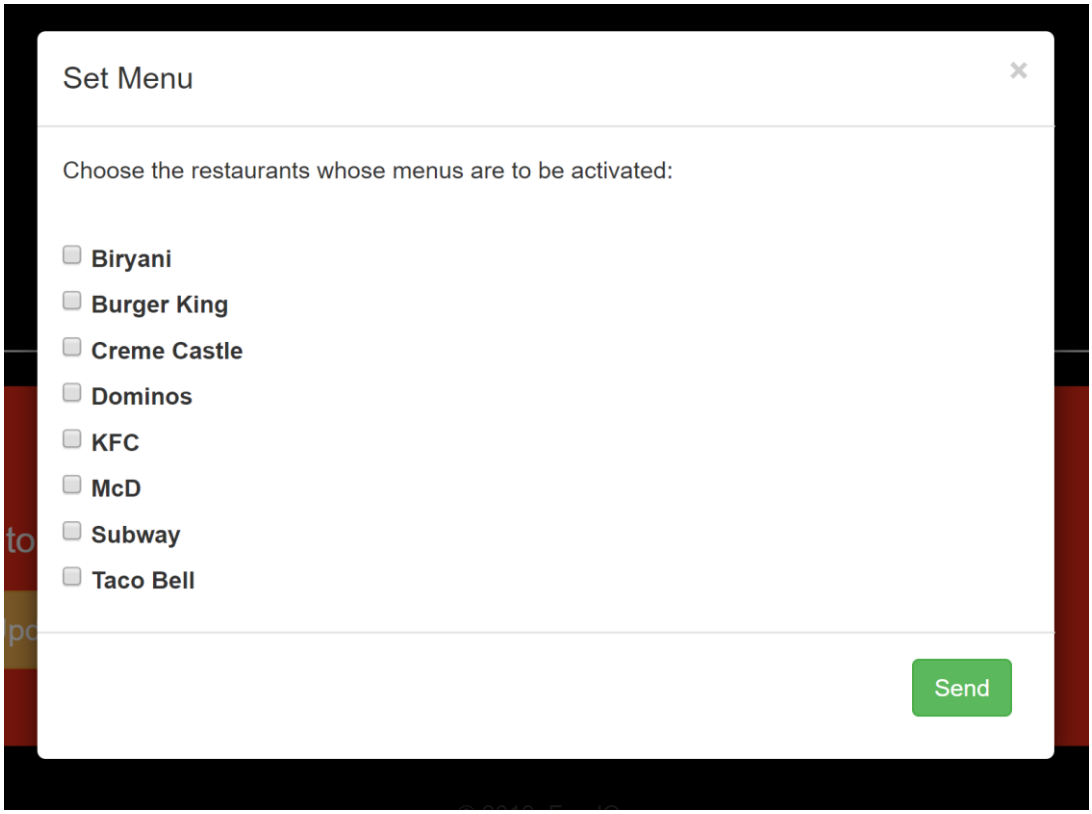
Admin Login



Admin Page



Sending update to all customers



Choosing which menus to activate

Division of Labour

➢ *Problem Statement*	ALL
➢ *Software Requirement Specification (SRS)*	Rebhav
➢ *Data Flow Diagram*	ALL
➢ *Use Case Diagram*	Kabir
➢ *Sequence Diagram*	Kabir
➢ *Class Diagram*	Daniel and Rebhav
➢ *Activity Diagram*	Daniel
➢ *State Chart Diagram*	Rebhav
➢ *User Side Website*	Daniel
➢ *Backend Set-up*	Daniel
➢ *Admin Side Website*	Kabir
➢ *Database Entries*	Rebhav
➢ *Report*	Kabir
➢ *Presentation*	Kabir

Design Challenges

This project taught us several valuable lessons. We faced quite a few challenges along the way. First off, we could have managed our time better and planned the project out in such a way that we should have worked consistently throughout the semester and not right at the end.

Working in a team also proved to be a task in itself. With everyone having different priorities and schedules and commitments, it becomes difficult to collaborate and build a project with loads of other stuff going on.

Initially, we were not using Github, so code collaboration was a bit messy. However, we learnt a lot from this and had some great takeaways for next time.

Future Path

We plan to develop this into a full-fledged working project that can actually be deployed by FoodOn, the start-up on campus.

There are a lot more features that could be incorporated, such as editing and adding menus on the website itself, adding a payment gateway, adding a reward points system, live order tracking, email/SMS updates and the option of sending a message to the admins.

Conclusion

We feel that even though we have worked extremely hard on this project, we still need to work on it before it can actually be deployed. However, we have gained a lot of knowledge in the process and are almost certain and confident that we will be able to do so by early next year. We learnt valuable lessons about working as a team and managing time. We also learnt about various new software and technologies and we are truly grateful for the same.

Project Link

https://github.com/KabirKanha/foodon-webapp

(Private at the moment)